

Sequences & Mathematical Induction

5.1 Sequences

5.2&3 Mathematical Induction (الاستقراء الرياضي)



Watch this lecture and download the slides



<http://jarrar-courses.blogspot.com/2014/03/discrete-mathematics-course.html>

More Online Courses at: <http://www.jarrar.info>

Acknowledgement:

This lecture is based on (but not limited to) to chapter 5 in “Discrete Mathematics with Applications by Susanna S. Epp (3rd Edition)”.

Sequences & Mathematical Induction

5.2&3 Mathematical Induction

In this lecture:



☐ Part 1: What is Mathematical Induction

- ☐ Part 2: Induction as a Method of Proof/Thinking
- ☐ Part 3: Proving *sum of integers* and *geometric sequences*
- ☐ Part 4: Proving a *Divisibility Property* and *Inequality*
- ☐ Part 5: Proving a *Property of a Sequence*
- ☐ Part 6: Induction Versus Deduction Thinking

What is Mathematical Induction

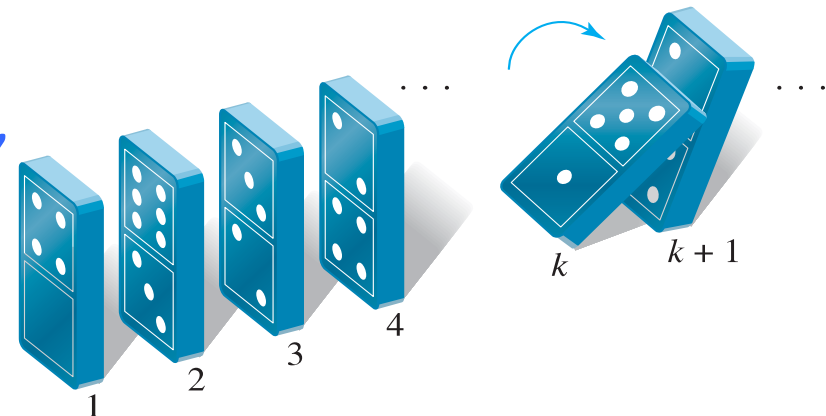
Mathematical induction is one of the more **recently developed methods of proof** in mathematics.

History:

The first use of mathematical induction was by الكرجي/Al-kraji (1000AD) in his book الفخري/ Al-Fakhri to prove math sequences. In 1883 Augustus De Morgan described it carefully and named mathematical induction.

The idea:

If the k^{th} domino falls backward, it pushes the $(k+1)^{\text{st}}$ domino backward.



What is Mathematical Induction

Principle of Mathematical Induction

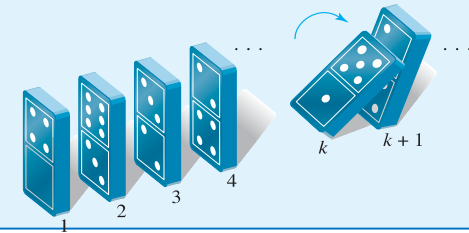
Let $P(n)$ be a property that is defined for integers n , and let a be a fixed integer. Suppose the following two statements are true:

1. $P(a)$ is true.
2. For all integers $k \geq a$, if $P(k)$ is true then $P(k + 1)$ is true.

Then the statement

for all integers $n \geq a$, $P(n)$

is true.



Example:

how to know whether this $P(n)$ can be true?

$P(n)$: For all integers $n \geq 8$, n cents can be obtained using 3¢ and 5¢ coins.

→ Moves from specific cases to create a general rule (conjecture/ حدس), this is why it is called **Principle, not a theorem**

What is Mathematical Induction

Example

How to know whether this statement can be true?

For all integers $n \geq 8$, n cents can be obtained using 3¢ and 5¢ coins.

For all integers $n \geq 8$, $P(n)$ is true,
where $P(n)$ is the sentence “ n cents
can be obtained using 3¢ and 5¢
coins.”

Then we need to prove that $P(n+1)$ is
also true

Number of Cents	How to Obtain It
8¢	3¢ + 5¢
9¢	3¢ + 3¢ + 3¢
10¢	5¢ + 5¢
11¢	3¢ + 3¢ + 5¢
12¢	3¢ + 3¢ + 3¢ + 3¢
13¢	3¢ + 5¢ + 5¢
14¢	3¢ + 3¢ + 3¢ + 5¢
15¢	5¢ + 5¢ + 5¢
16¢	3¢ + 3¢ + 5¢ + 5¢
17¢	3¢ + 3¢ + 3¢ + 3¢ + 5¢

Sequences & Mathematical Induction

5.2&3 Mathematical Induction

In this lecture:

☐ Part 1: *What is Mathematical Induction*



☒ **Part 2: Induction as a Method of Proof/Thinking**

☐ Part 3: **Proving** *sum of integers and geometric sequences*

☐ Part 4: **Proving** *a Divisibility Property and Inequality*

☐ Part 5: **Proving** *a Property of a Sequence*

☐ Part 6: Induction Versus Deduction Thinking

Mathematical Induction as a Method of Proof

Proving a statement by mathematical induction is a two-step process. The first step is called the *basis step*, and the second step is called the *inductive step*.

Method of Proof by Mathematical Induction

Consider a statement of the form, “For all integers $n \geq a$, a property $P(n)$ is true.” To prove such a statement, perform the following two steps:

Step 1 (basis step): Show that $P(a)$ is true.

Step 2 (inductive step): Show that for all integers $k \geq a$, if $P(k)$ is true then $P(k + 1)$ is true. To perform this step,

suppose that $P(k)$ is true, where k is any particular but arbitrarily chosen integer with $k \geq a$.

*[This supposition is called the **inductive hypothesis**.]*

Then

show that $P(k + 1)$ is true.

Mathematical Induction as a Method of Proof

Example

How to know whether this statement can be true?

For all integers $n \geq 8$, n cents can be obtained using 3¢ and 5¢ coins.

Let the property $P(n)$ be the sentence: n ¢ can be obtained using 3¢ and 5¢ coins. $\leftarrow P(n)$

Step 1 (basis step): Show $P(8)$ is true: $P(8)$ is true as 8¢ obtained by one 3¢ and one 5¢

Step 2(inductive step): Show for all integers $k \geq 8$, if $P(k)$ is true then $P(k+1)$ is true:

[Suppose that $P(k)$ is true for a particular but arbitrarily chosen integer $k \geq 8$. That is:]

Suppose k is any integer $k \geq 8$, k ¢ obtained by 3¢ and 5¢ . $\leftarrow P(k)$ **inductive hypothesis**

[We must show that $P(k + 1)$ is true. That is:] We must show that

$(k + 1)$ ¢ / can be obtained using 3 ¢ / and 5 ¢ / coins. $\leftarrow P(k + 1)$

Case 1 (There is a 5¢ coin among those used to make up the k ¢):

replace the 5c/ coin by two 3c/ coins; the result will be $(k + 1)$ c/.

Case 2 (There is not a 5¢ coin among those used to make up the k ¢):

because $k \geq 8$, at least three 3¢ must have been used. So remove three 3¢ and replace them by two 5¢; the result will be $(k + 1)$ ¢.

Thus in either case $(k + 1)$ ¢ can be obtained using 3¢ and 5¢ [as was to be shown].

Sequences & Mathematical Induction

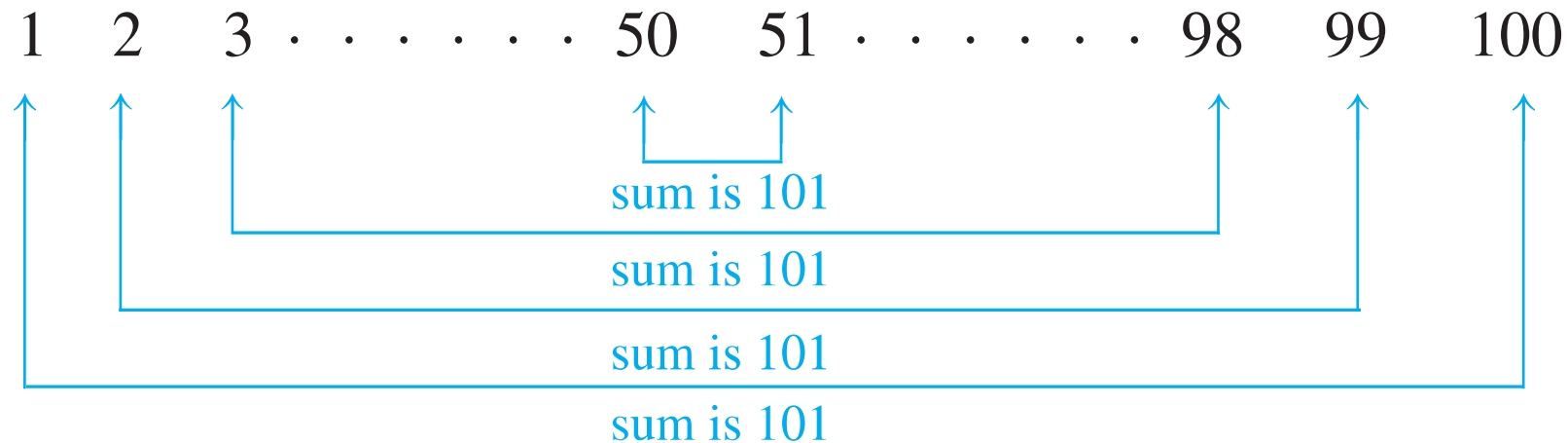
5.2&3 Mathematical Induction

In this lecture:

- ☐ Part 1: What is Mathematical Induction
- ☐ Part 2 : Induction as a Method of Proof/Thinking
-  ☐ Part 3: **Proving Sum of Integers and Geometric Sequences**
- ☐ Part 4: Proving a *Divisibility Property and Inequality*
- ☐ Part 5: Proving a *Property of a Sequence*
- ☐ Part 6: Induction Versus Deduction Thinking

Sum of the First n Integers

Who can sum all numbers from 1 to 100?



$$1 + 2 + 3 + \cdots + n = \frac{n(n + 1)}{2}$$

Theorem 5.2.2 Sum of the First n Integers

For all integers $n \geq 1$, $1 + 2 + 3 + \dots + n = \frac{n(n+1)}{2}$

Same Question: Prove that these programs prints the same results in case $n \geq 1$

```
For (i=1, i≤n; i++)
```

```
    S=S+i;
```

```
Print ("%d", S);
```

```
S=(n(n+1))/2
```

```
Print ("%d",S);
```

Theorem 5.2.2 Sum of the First n Integers

For all integers $n \geq 1$, $1 + 2 + 3 + \dots + n = \frac{n(n+1)}{2}$

Same Question: Prove that these programs prints the same results in case $n \geq 1$

For (i=1, i≤n; i++)

S=S+i;

Print ("%d", S);

S=(n(n+1))/2

Print ("%d", S);

Proving that both programs produce the same results is like proving that:

$$1 + 2 + 3 + \dots + n = \frac{n(n+1)}{2} \quad \leftarrow P(n)$$

Basis Step: Show that $P(1)$ is true. $P(1)$: $1 = 1(1+1)/2 = 1$ Thus $P(1)$ is true

Inductive Step: Show that for all integers $k \geq 1$, if $P(k)$ is true then $P(k+1)$ is also true:

Suppose: $1+2+3+\dots+k = \frac{k(k+1)}{2}$ is true $\leftarrow P(k)$ inductive hypothesis

$$P(k+1) = 1+2+\dots+k + (k+1) = \frac{(k+1)(k+2)}{2} \quad \leftarrow P(k+1)$$
$$= P(k) + (k+1)$$

$$= \frac{k(k+1)}{2} + (k+1) = \frac{k(k+1)}{2} + \frac{2(k+1)}{2}$$

$$= \frac{k^2+k}{2} + \frac{2(k+1)}{2} = \frac{k^2+3k+2}{2}$$

$$= \frac{(k+1)(k+2)}{2}$$

Same

Examples of Sums

Evaluate $2 + 4 + 6 + \cdots + 500$.

$$\begin{aligned} 2 + 4 + 6 + \cdots + 500 &= 2 \cdot (1 + 2 + 3 + \cdots + 250) \\ &= 2 \cdot \left(\frac{250 \cdot 251}{2} \right) \\ &= 62,750. \end{aligned}$$

Evaluate $5 + 6 + 7 + 8 + \cdots + 50$.

$$\begin{aligned} 5 + 6 + 7 + 8 + \cdots + 50 &= (1 + 2 + 3 + \cdots + 50) - (1 + 2 + 3 + 4) \\ &= \frac{50 \cdot 51}{2} - 10 \\ &= 1,265 \end{aligned}$$

For an integer $h \geq 2$, write $1 + 2 + 3 + \cdots + (h-1)$ in closed form.

$$\begin{aligned} 1 + 2 + 3 + \cdots + (h-1) &= \frac{(h-1) \cdot [(h-1) + 1]}{2} \\ &= \frac{(h-1) \cdot h}{2} \end{aligned}$$

Theorem 5.2.3 Sum of a Geometric Sequence

For any real number r except 1, and any integer $n \geq 0$,

$$\sum_{i=0}^n r^i = \frac{r^{n+1} - 1}{r - 1}.$$

Proof (by mathematical induction):

$$\sum_{i=0}^0 r^i = \frac{r^{0+1} - 1}{r - 1} \leftarrow P(0) = \frac{r - 1}{r - 1} = 1$$

$$\sum_{i=0}^k r^i = \frac{r^{k+1} - 1}{r - 1} \leftarrow P(k) \text{ inductive hypothesis}$$

$$\begin{aligned} \sum_{i=0}^{k+1} r^i &= \frac{r^{k+2} - 1}{r - 1} \leftarrow P(k+1) \\ &= \sum_{i=0}^k r^i + r^{k+1} \\ &= \frac{r^{k+1} - 1}{r - 1} + r^{k+1} \\ &= \frac{r^{k+1} - 1}{r - 1} + \frac{r^{k+1}(r - 1)}{r - 1} \\ &= \frac{(r^{k+1} - 1) + r^{k+1}(r - 1)}{r - 1} \\ &= \frac{r^{k+1} - 1 + r^{k+2} - r^{k+1}}{r - 1} \\ &= \frac{r^{k+2} - 1}{r - 1} \end{aligned}$$

Mathematics in Programming

Example : Finding the sum of a geometric series

Prove that these codes will return the same output.

n.

```
int n, r, sum=0;
int i;
scanf("%d",&n);
scanf("%d",&r);

if(r != 1) {
    for(i=0 ; i<=n ; i++) {
        sum = sum + pow(r,i);
    }
    printf("%d\n", sum);
}
```

```
int n, r, sum=0;
scanf("%d",&n);
scanf("%d",&r);

if(r != 1) {
    sum=((pow(r,n+1))-1)/(r-1);
    printf("%d\n", sum);
}
```

This code is proposed by a student/Zaina!

Examples of Sums of a Geometric Sequence

In each of (a) and (b) below, assume that m is an integer that is greater than or equal to 3. Write each of the sums in closed form.

(a) $1+3+3^2+\cdots+3^{m-2}$

$$\begin{aligned} 1 + 3 + 3^2 + \cdots + 3^{m-2} &= \frac{3^{(m-2)+1} - 1}{3 - 1} \\ &= \frac{3^{m-1} - 1}{2}. \end{aligned}$$

(b) $3^2+3^3+3^4+\cdots+3^m$

$$\begin{aligned} 3^2 + 3^3 + 3^4 + \cdots + 3^m &= 3^2 \cdot (1 + 3 + 3^2 + \cdots + 3^{m-2}) \\ &= 9 \cdot \left(\frac{3^{m-1} - 1}{2} \right) \end{aligned}$$

Sequences & Mathematical Induction

5.2&3 **Mathematical Induction**

In this lecture:

- ☐ Part 1: What is Mathematical Induction
- ☐ Part 2 : Induction as a Method of Proof/Thinking
- ☐ Part 3: Proving Sum of Integers and Geometric Sequences
-  ☐ Part 4: **Proving a Divisibility Property and Inequality**
- ☐ Part 5: Proving a *Property of a Sequence*
- ☐ Part 6: Induction Versus Deduction Thinking

Proposition 5.3.1 Proving a Divisibility Property

For all integers $n \geq 0$, $2^{2n} - 1$ is divisible by 3.

Proof (by mathematical induction):

$$3 \mid 2^{2n} - 1 \quad \leftarrow P(n)$$

Basis Step: *Show that $P(0)$ is true.*

$$P(0): 2^{2 \cdot 0} - 1 = 2^0 - 1 = 1 - 1 = 0 \quad \text{As } 3 \mid 0, \text{ thus } P(0) \text{ is true.}$$

Inductive Step: *Show that for all integers $k \geq 0$, if $P(k)$ is true then $P(k + 1)$ is also true:*

Suppose: $2^{2k} - 1$ is divisible by 3. $\leftarrow P(k)$ inductive hypothesis

$$2^{2k} - 1 = 3r \text{ for some integer } r.$$

$$2^{2(k+1)} - 1 \text{ is divisible by 3. } \leftarrow P(k+1)$$

$$2^{2(k+1)} - 1 = 2^{2k+2} - 1 \quad \text{by the laws of exponents}$$

$$= 2^{2k} \cdot 2^2 - 1 = 2^{2k} \cdot 4 - 1$$

$$= 2^{2k}(3 + 1) - 1 = 2^{2k} \cdot 3 + (2^{2k} - 1) = 2^{2k} \cdot 3 + 3r$$

$$= 3(2^{2k} + r) \quad \text{Which is integer}$$

so, by definition of divisibility, $2^{2(k+1)} - 1$ is divisible by 3

Mathematics in Programming

Example : Proving Property of a Sequence

What will the output of this program be for any input n?

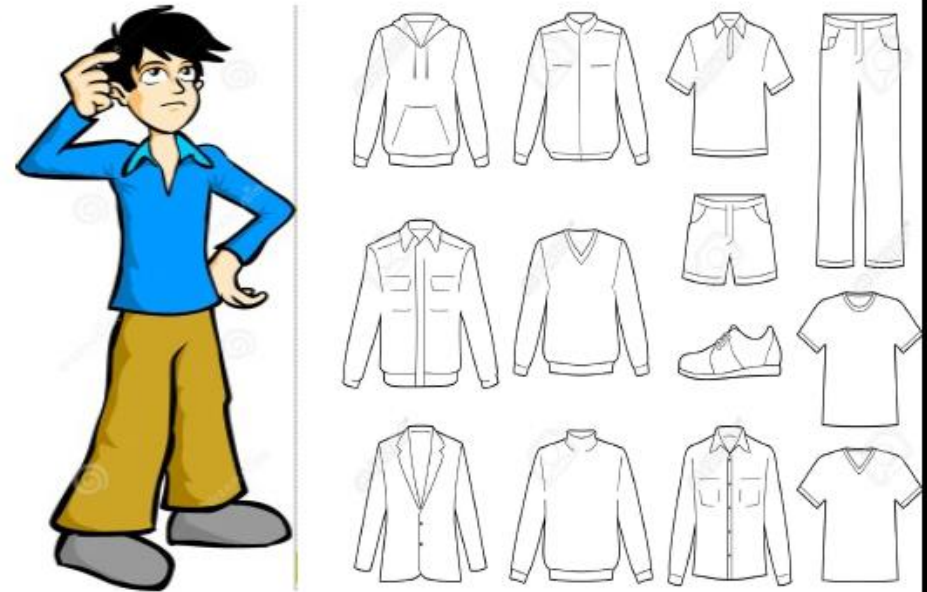
```
int n;  
scanf("%d",&n);  
  
if(n >= 0) {  
    if( (pow(2,(2*n)) - 1) %3 == 0)    \\ does  $2^{2n}-1 \mid 3$ ?? \\  
        printf("this property is true");  
    else  
        printf("this property isn't true");  
}
```

Mathematics in Programming

Example : Proving Property of a Sequence

What will this guy choose to wear today ?
(What is the output of the program)

```
int x, y;  
scanf("%d %d", &x, &y);  
if(x%2 == 0)  
    x=x+1;  
if(pow(x, 2)%2 != 0)  
    printf("White Shirt");  
else  
    printf("Black Shirt");  
  
if((pow(7, y)-1)%6==0)  
    printf("Black boot");  
else  
    printf ("White Boot");
```



Proposition 5.3.2 Proving Inequality

For all integers $n \geq 3$, $2n + 1 < 2^n$.

Proof (by mathematical induction):

Let $P(n)$ be $2n+1 < 2^n$

Basis Step: *Show that $P(3)$ is true.* $P(3)$: $2 \cdot 3 + 1 < 2^3$ which is true.

Inductive Step: *Show that for all integers $k \geq 3$, if $P(k)$ is true then $P(k + 1)$ is also true:*

Suppose: $2k + 1 < 2^k$ is true $\leftarrow P(k)$ inductive hypothesis

$$2(k+1) + 1 < 2^{k+1} \quad \leftarrow P(k+1)$$

$$2k+3 = (2k+1) + 2 \quad \text{by algebra}$$

$$< 2^k + 2^k \quad \text{as } 2k + 1 < 2^k \text{ by the inductive hypothesis}$$

$$\therefore 2k + 3 < 2 \cdot 2^k = 2^{k+1} \quad \text{and because } 2 < 2^k \quad (k \geq 2)$$

[This is what we needed to show.]

Sequences & Mathematical Induction

5.2&3 **Mathematical Induction**

In this lecture:

- ☐ Part 1: What is Mathematical Induction
- ☐ Part 2 : Induction as a Method of Proof/Thinking
- ☐ Part 3: Proving Sum of Integers and Geometric Sequences
- ☐ Part 4: Proving a Divisibility Property and Inequality
-  ☐ Part 5: **Proving a Property of a Sequence**
- ☐ Part 6: Induction Versus Deduction Thinking

Proving a Property of a Sequence

Example

Define a sequence $a_1, a_2, a_3 \dots$ as follows:

$$a_1 = 2$$

$$a_k = 5a_{k-1} \quad \text{for all integers } k \geq 2.$$

Write the first four terms of the sequence.

$$a_1 = 2$$

$$a_2 = 5a_{2-1} = 5a_1 = 5 \cdot 2 = 10$$

$$a_3 = 5a_{3-1} = 5a_2 = 5 \cdot 10 = 50$$

$$a_4 = 5a_{4-1} = 5a_3 = 5 \cdot 50 = 250$$

→ The terms of the sequence satisfy the equation $a_n = 2 \cdot 5^{n-1}$

Proving a Property of a Sequence

Example

Prove this property:

$$a_n = 2 \cdot 5^{n-1} \text{ for all integers } n \geq 1$$

Basis Step: *Show that $P(1)$ is true.*

$$a_1 = 2 \cdot 5^{1-1} - 1 = 2 \cdot 5^0 - 1 = 2$$

Inductive Step: *Show that for all integers $k \geq 1$, if $P(k)$ is true then $P(k+1)$ is also true:*

$$\text{Suppose: } a_k = 2 \cdot 5^{k-1}$$

$\leftarrow P(k)$ inductive hypothesis

$$a_{k+1} = 2 \cdot 5^k$$

$\leftarrow P(k+1)$

$$= 5a_{(k+1)-1}$$

by definition of $a_1, a_2, a_3 \dots$

$$= 5a_k$$

$$= 5 \cdot (2 \cdot 5^{k-1})$$

by the hypothesis

$$= 2 \cdot (5 \cdot 5^{k-1})$$

$$= 2 \cdot 5^k$$

[This is what we needed to show.]

Sequences & Mathematical Induction

5.2&3 Mathematical Induction

In this lecture:

- ☐ Part 1: What is Mathematical Induction
- ☐ Part 2 : Induction as a Method of Proof/Thinking
- ☐ Part 3: Proving Sum of Integers and Geometric Sequences
- ☐ Part 4: Proving a Divisibility Property and Inequality
- ☐ Part 5: Proving a Property of a Sequence
-  ☐ Part 6: **Induction Versus Deduction Thinking**

Induction Versus Deduction Reasoning

Deduction Reasoning

If Every man is person and
Sami is Man,
then Sami is Person

If my highest mark this
semester is 82%, then my
average will not be more than
82%

Induction Reasoning

For all integers $n \geq 8$, n
cents can be obtained
using 3¢ and 5¢ coins.

We had a quiz each lecture
in the past months, so we
will have a quiz next lecture

Induction Versus Deduction Reasoning

Deduction Reasoning

Based on facts, definitions, ,
theorems, laws

Moves from general
observation to specific results

Provides proofs

Induction Reasoning

Based on observation,
past experience, patterns

Moves from specific cases
to create a general rule

Provides conjecture/حدس

More slides

More slides from students

Student: Ehab, 2016

Not reviewed or verified

Example¹

prove the following property:

for all integers $n \geq 1$, $1 \times 2 + 2 \times 3 + 3 \times 4 + \dots + (n)(n+1) = \frac{(n)(n+1)(n+2)}{3}$

basis step : show $p(1)$ is true.

$$P(1): 1 \times 2 = \frac{(1)(2)(3)}{3}$$

left-hand side is $1 \times 2 = 2$

right-hand side is $\frac{(1)(2)(3)}{3} = 2$

thus $p(1)$ is true

inductive step : Show that for all integers $k \geq 1$, if $P(k)$ is true then $P(k + 1)$ is also true:

suppose that $p(k)$ is true

$$p(k) = 1 \times 2 + 2 \times 3 + 3 \times 4 + \dots + (k)(k+1) = \frac{(k)(k+1)(k+2)}{3} \quad \leftarrow P(k) \text{ inductive hypothesis}$$

$$p(k+1) = 1 \times 2 + 2 \times 3 + 3 \times 4 + \dots + (k)(k+1) + (k+1)((k+1)+1)$$

$$= [1 \times 2 + 2 \times 3 + 3 \times 4 + \dots + (k)(k+1)] + (k+1)((k+1)+1)$$

$$= \frac{(k)(k+1)(k+2)}{3} + (k+1)(k+2)$$

$$= \frac{(k)(k+1)(k+2)}{3} + \frac{3(k+1)(k+2)}{3}$$

$$= \frac{(k+1)(k+2)(k+3)}{3} = \text{right side} \quad \text{[This is what we needed to show.]}$$

Then $p(k)$ works for all $n \geq 1$.

¹ CALCULUS with Analytic Geometry, Earl W. Swokowski

Example¹

Show that For any integer $n \geq 5$, $4n < 2^n$.

basis step : show $P(n = 5)$ is true.

$$4n = 4 \times 5 = 20, \text{ and } 2^n = 2^5 = 32.$$

Since $20 < 32$, thus $p(n=5)$ is true

inductive step : Show that for all integers $k > 0$, if $p(k)$ is true then $p(k+1)$ is true:

suppose $p(k)$ is true for $k \geq 5$ $\leftarrow$ **$P(k)$ inductive hypothesis**

$$p(k+1): \quad 4(k+1) = 4k + 4, \text{ and, by assumption } [4k] + 4 < [2^k] + 4$$

Since $k \geq 5$, then $4 < 32 \leq 2^k$. Then we get

$$2^k + 4 < 2^k + 2^k =$$

$$= 2 \times 2^k$$

$$= 2^1 \times 2^k$$

$$= 2^{k+1}$$

Then $4(k+1) < 2^{k+1}$, hence $p(k+1)$ is true *[This is what we needed to show.]*

¹ question taken from this book: CALCULUS with Analytic Geometry, Earl W. Swokowski

Example¹

show that For all $n \geq 1$, $8^n - 3^n$ is divisible by 5.

basis step : show that $p(1)$ is true

$$8^1 - 3^1 =$$

$$= 8 - 3$$

$= 5$ which is clearly divisible by 5.

inductive step : Show that for all integers $k > 0$, if $p(k)$ is true then $p(k+1)$ is true:

Suppose $p(k)$ is true ($8^k - 3^k$ is divisible by 5) $\leftarrow P(k)$ **inductive hypothesis**

$$8^{k+1} - 3^{k+1} =$$

$$= 8^{k+1} - 3 \times 8^k + 3 \times 8^k - 3^{k+1}$$

$$= 8^k(8 - 3) + 3(8^k - 3^k)$$

$$= 8^k(5) + 3(8^k - 3^k)$$

The first term in $8^k(5) + 3(8^k - 3^k)$ has 5 as a factor (explicitly), and the second term is divisible by 5 (by assumption). Since we can factor a 5 out of both terms, then the entire expression,

$8^k(5) + 3(8^k - 3^k) = 8^{k+1} - 3^{k+1}$, must be divisible by 5.

[This is what we needed to show.]

¹ question taken from this book: CALCULUS with Analytic Geometry, Earl W. Swokowski

Example¹

$1^3 + 2^3 + 3^3 + \dots + n^3 = \frac{n^2 (n + 1)^2}{4}$. show that this equation is true for all integers $n \geq 1$.

Basis step: show that $p(1)$ is true.

$$\text{Left Side} = 1^3 = 1$$

$$\text{Right Side} = \frac{1^2 (1 + 1)^2}{4} = 1$$

hence $p(1)$ is true.

Inductive step: Show that for all integers $k > 0$, if $p(k)$ is true then $p(k+1)$ is true:

suppose that $p(k)$ is true $\leftarrow P(k)$ inductive hypothesis

$$1^3 + 2^3 + 3^3 + \dots + k^3 + (k + 1)^3$$

$$= \frac{k^2 (k + 1)^2}{4} + (k+1)^3$$

$$= \frac{k^2 (k + 1)^2}{4} + \frac{4(k+1)^3}{4}$$

$$= \frac{(k + 1)^2 [k^2 + 4k + 4]}{4}$$

$$= \frac{(k + 1)^2 [(k + 2)^2]}{4}$$

= right side

[This is what we needed to show.]

¹ question taken from this book: CALCULUS with Analytic Geometry, Earl W. Swokowski

Example¹

Prove that for any integer number $n \geq 1$, $n^3 + 2n$ is divisible by 3

Basis Step: show that $p(1)$ is true.

Let $n = 1$ and calculate $n^3 + 2n$

$$1^3 + 2(1) = 3$$

3 is divisible by 3, hence $p(1)$ is true.

Inductive Step: Show that for all integers $k > 0$, if $p(k)$ is true then $p(k+1)$ is true:

suppose that $p(k)$ is true $\leftarrow$ **$P(k)$ inductive hypothesis**

$$(k+1)^3 + 2(k+1)$$

$$= k^3 + 3k^2 + 5k + 3$$

$$= [k^3 + 2k] + [3k^2 + 3k + 3]$$

$$= 3[k^3 + 2k] + 3[k^2 + k + 1]$$

$$= 3[[k^3 + 2k] + k^2 + k + 1]$$

Hence $(k+1)^3 + 2(k+1)$ is also divisible by 3 and therefore statement $P(k+1)$ is true.

[This is what we needed to show.]

¹ question taken from this book: CALCULUS with Analytic Geometry, Earl W. Swokowski